Yitzchak Shmalo

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Research interests

Probability and random matrices; mathematical foundations of learning; reliable AI-assisted research. Shared memory, retrieval, verification and coordination for agents working on mathematical and scientific problems.

Appointments

2025–present Postdoctoral Fellow, Einstein Institute of Mathematics, The Hebrew University of Jerusalem. Hosts: Sasha Sodin and Amit Daniely.

Summer 2024 Research Intern, NASA Jet Propulsion Laboratory, California Institute of Technology. Uncertainty Quantification and Statistical Analysis group. Mentors: Otto Lamminpää and Jouni Susiluoto. Radiative-transfer emulation, low-rank and sparse kernels, and random-matrix pruning.

2019–2025 Graduate Teaching and Research Assistant, Department of Mathematics, Pennsylvania State University.

Education

Ph.D. in Mathematics, Pennsylvania State University, 2025. Advisor: Leonid Berlyand. Dissertation: Applications of Random Matrix Theory to Deep Learning.

M.A. in Mathematics, Yeshiva University, 2017. Advisor: Marian Gidea.
B.A. in Mathematics, Yeshiva University, 2016.

Peer reviewed publications

1. Y. Shmalo. Extreme least singular values of Gaussian row submatrices and a phase retrieval stability problem. Bulletin of the Australian Mathematical Society, First View, 1–9. Published online 14 October 2026. DOI: 10.1017/S0004972726101713.

2. L. Berlyand, E. Sandier, Y. Shmalo and L. Zhang. Enhancing accuracy in deep learning using random matrix theory. Journal of Machine Learning 3 (2024), no. 4, 347–412. DOI: 10.4208/jml.231220.

3. Y. Shmalo. Stability of accuracy for the training of DNNs via the uniform doubling condition. Annals of Mathematics and Artificial Intelligence 92 (2024), no. 2, 439–483. DOI: 10.1007/s10472-023-09919-1.

4. E. Prodan and Y. Shmalo. The K-theoretic bulk-boundary principle for dynamically patterned resonators. Journal of Geometry and Physics 135 (2019), 135–171. DOI: 10.1016/j.geomphys.2018.10.005.

5. M. Gidea and Y. Shmalo. Combinatorial approach to detection of fixed points, periodic orbits, and symbolic dynamics. Discrete and Continuous Dynamical Systems 38 (2018), no. 12, 6123–6148.

Selected public preprints

The works below are preprints, separate from the refereed publications above. Manuscripts, source code and author contribution statements are linked where available.

1. Y. Shmalo. Math Brain: shared, auditable memory for multi-agent mathematical research. 2026. Code, public snapshot and experimental record.

2. L. Berlyand, T. Bourdais, H. Owhadi and Y. Shmalo. Pruning deep neural networks via the Marchenko–Pastur distribution. 2026. arXiv:2606.02608.

3. Y. Shmalo. Neural means and kernel corrections for operator learning. 2026. arXiv:2609.00389.

4. L. Berlyand, R. Sarapin, Y. Shmalo, V. Slavin and S. Sodin. Randomly initialized autoencoders: fixed points and edge-of-chaos. 2026. arXiv:2608.14638.

5. Y. Shmalo. The smallest eigenvalue of the neural tangent kernel of a deep tanh network in the ordered phase. 2026. Manuscript and numerical scripts.

6. Y. Shmalo. A law of robustness for two-layer neural networks with arbitrary weights. 2026. arXiv:2607.07778.

7. Y. Shmalo. Mesoscopic rectangular spikes under subspace local laws: outlier values and singular subspaces. 2026. arXiv:2608.19594.

8. Y. Shmalo. The storage capacity of the Ising perceptron: verification of the outstanding numerical conditions. 2026. arXiv:2607.11010. Proof programs and certificates.

9. Y. Shmalo. Entropy and singularity of spectral and pivotal sets in planar percolation. 2026. arXiv:2609.15388.

10. Y. Shmalo. Fourier entropy of threshold functions and their compositions. 2026. arXiv:2610.16069.

11. Y. Shmalo. Gaussian-integer spectra of random sign matrices. 2026. Manuscript and exact-arithmetic verification.

12. Y. Shmalo. Finite-state counterexamples to Ross's second queueing conjecture. 2026. arXiv:2609.25084.

13. Y. Shmalo, J. Jenkins and O. Krupchytskyi. Deep learning weight pruning with RMT-SVD: increasing accuracy and reducing overfitting. 2023. arXiv:2303.08986.

Manuscript in preparation

W. Krauth and Y. Shmalo. Path convergence under identical diffusion noise. 2026. Finite-pattern diffusion paths and their limiting fluctuations under a common Brownian driver. Not yet publicly released.

Selected research software and projects

Math Brain

Public repository | Paper | Experimental methods and results

Designed the research program; directs the AI agents developing and testing a shared, auditable mathematical memory. The system keeps sourced results, hypotheses, computations and failed approaches as versioned cards with typed links. Work includes retrieval, corrections, tests of additions, and coordination across research sessions.

The 1 November 2026 public snapshot has **13,344 cards and 23,518 links**, including 231 retracted links. A 988-item Claude study compared recall and hooks with repository tools under a 600-second limit: 95.4% versus 82.0% accuracy, with different access interfaces. Among 806 items both finished, accuracy was 99.3% versus 99.4%. A separate files-plus-recall study scored 93.7% versus 91.4% on 1,726 reached Codex pairs, with longer runtime. Methods and negative results are linked above.

Computer assisted mathematical verification

Ising perceptron capacity. Interval-arithmetic proof programs and certificates for outstanding zero-margin capacity conditions, with Lean 4 checks of part of the certificate layer. AI systems developed most of the architecture and code under my supervision. External reproduction and peer review remain outstanding.

Gaussian-integer spectra. A construction contradicting Vu's Conjecture 9.5 for nonsymmetric sign matrices, with exact characteristic-polynomial checks in dimensions one through four. Matching upper bounds and the symmetric-matrix question remain open.

Cantor-set harmonic measures. Preprint on Lyons and Peres Question 16.7, with rational computations and selected Lean checks. Other software includes probability proof components and random-matrix pruning code.

Teaching

Pennsylvania State University, 2019–2025. Undergraduate mathematics teaching. Teaching Excellence Recognition, 2022.

Yeshiva University, 2017–2019. Graduate instructor in multivariable calculus, discrete structures and other courses. Teaching Excellence, 2017.

Yeshiva Rabbi Samson Raphael Hirsch, 2017–2019. Algebra I and II, Geometry, and curriculum design.

Honors and awards

2022 Teaching Excellence Recognition, Pennsylvania State University.
2020 Troxell Scholarship in Mathematics, Pennsylvania State University.
2019 Pettit Scholarship in Science, Pennsylvania State University.
2017 Teaching Excellence, Yeshiva University.
2016 Joseph Gunner Memorial Award for Excellence in Mathematics and Fisch Award, Yeshiva University.

Technical skills

Python, NumPy, SciPy, PyTorch; Mathematica, MATLAB, R and LaTeX. Arb interval arithmetic; Lean 4 with AI assistance. Design, supervision and evaluation of research workflows using Claude and Codex agents.